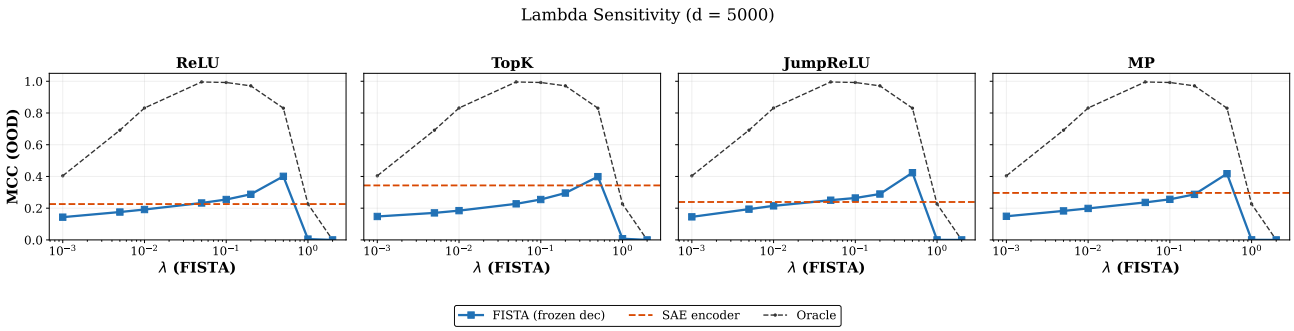

Figure 30: **Lambda sensitivity** ($d_z = 5000$)**.** Blue: FISTA on frozen SAE decoder across λ values. Orange dashed: SAE encoder (lambda-independent). Gray dashed: oracle FISTA. The frozen decoder peaks modestly above the SAE at $\lambda \approx 0.5$, but the gap to oracle persists at every λ.

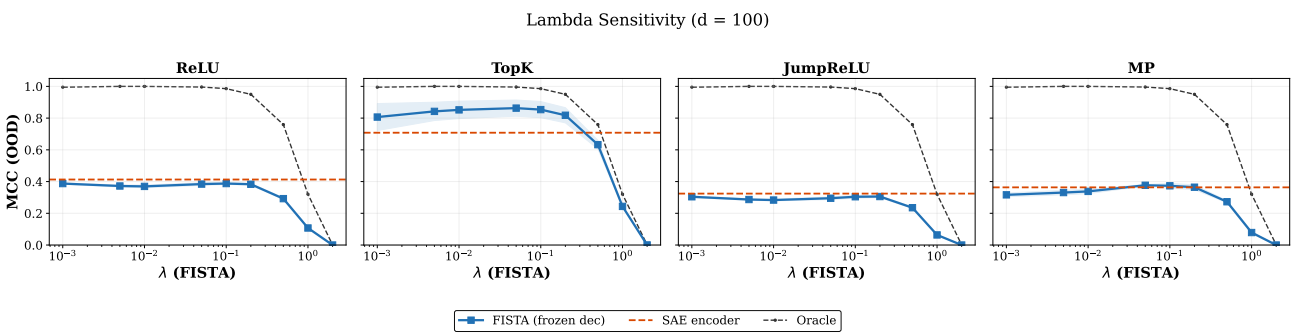

Figure 31: Lambda sensitivity, $d_z = 100$.

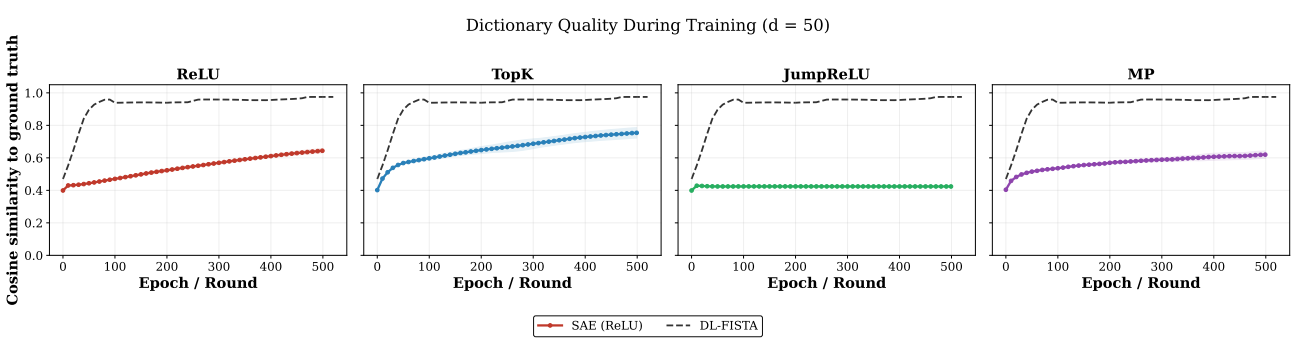

Figure 32: Dictionary quality during training, $d_z = 50$.

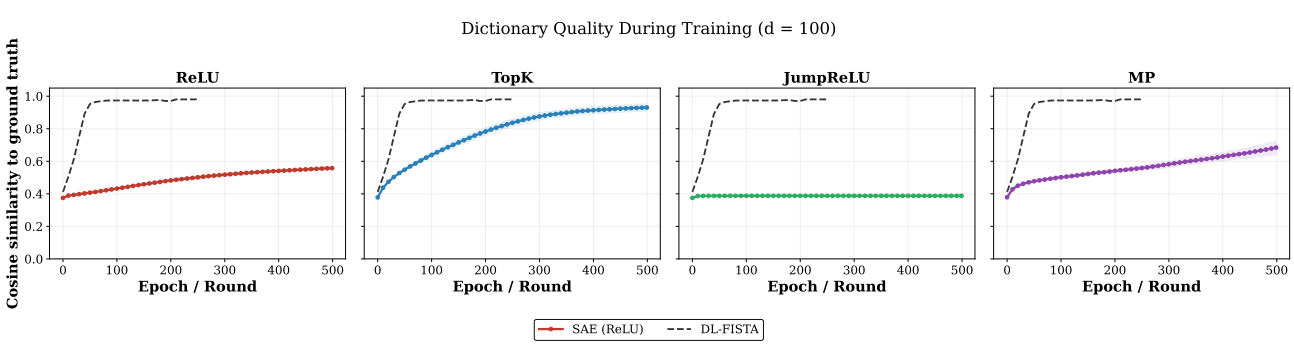

Figure 33: Dictionary quality during training, $d_z = 100$. At this scale, DL-FISTA converges to high cosine while SAEs plateau at lower values.

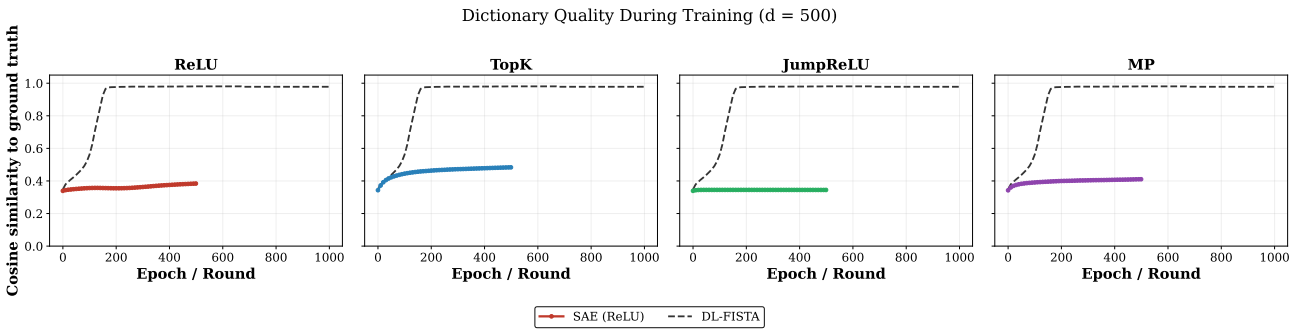

Figure 34: Dictionary quality during training, $d_z = 500$.

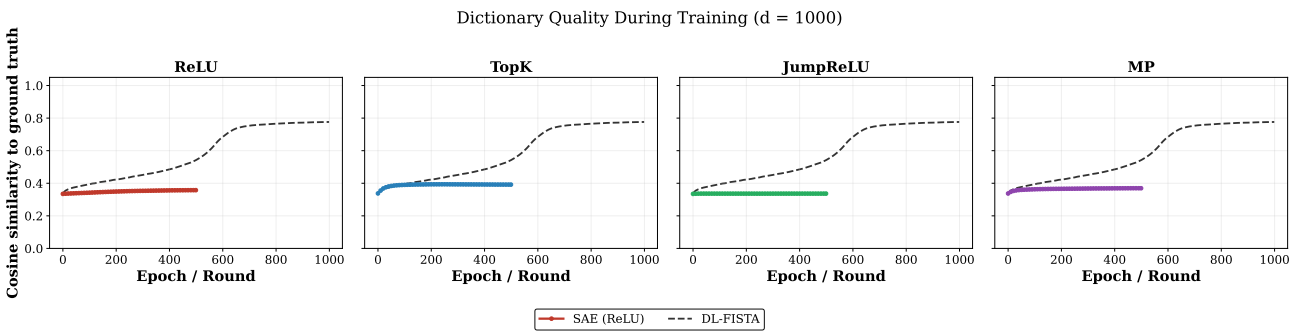

Figure 35: Dictionary quality during training, $d_z = 1000$.

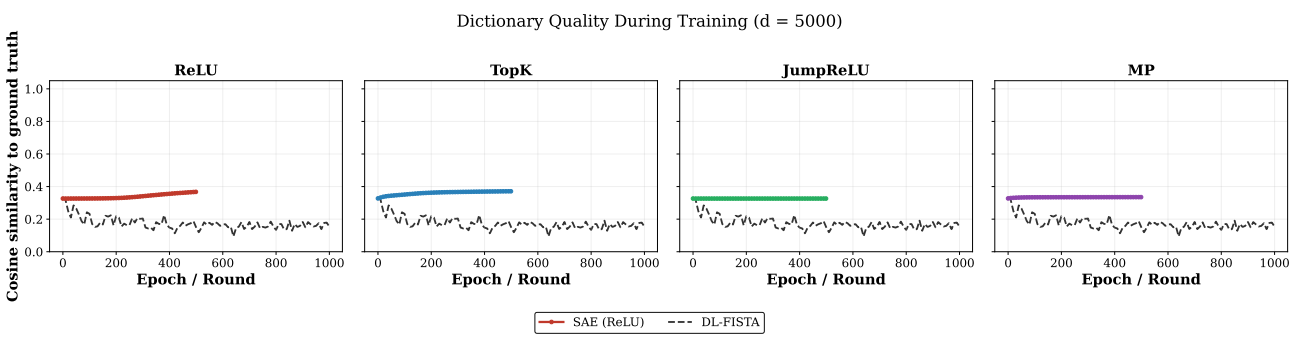

Figure 36: Dictionary quality during training, $d_z = 5000$. Neither SAEs nor DL-FISTA converge to high cosine at this scale.

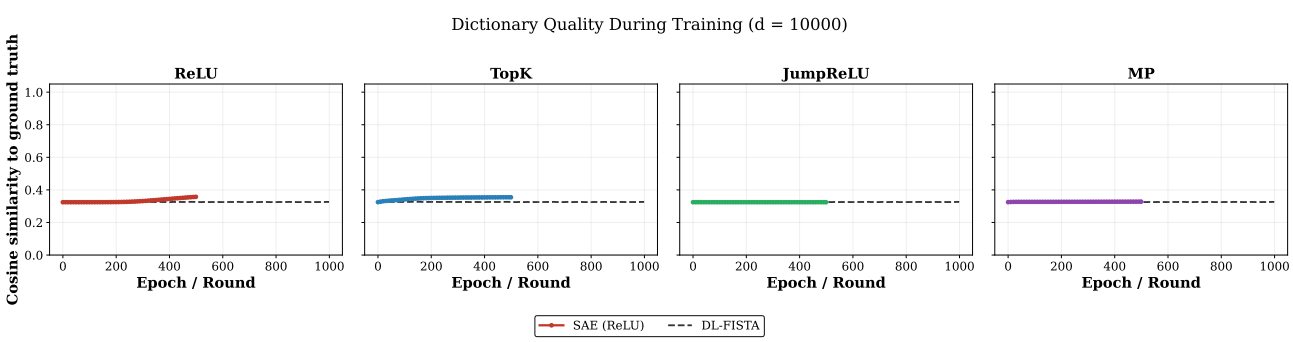

Figure 37: Dictionary quality during training, $d_z = 10000$.

Decoder warm-start convergence. The SAE decoder provides a consistent head start for DL-FISTA across all d_z. The advantage is largest at small d_z (where the SAE decoder is closer to the true dictionary) and diminishes at large d_z (where both initialisations require many rounds to converge).

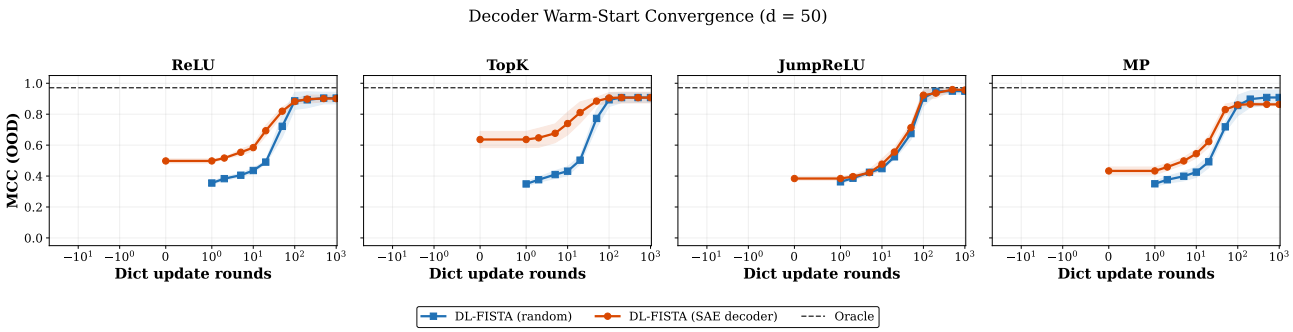

Figure 38: Decoder warm-start convergence, $d_z = 50$.

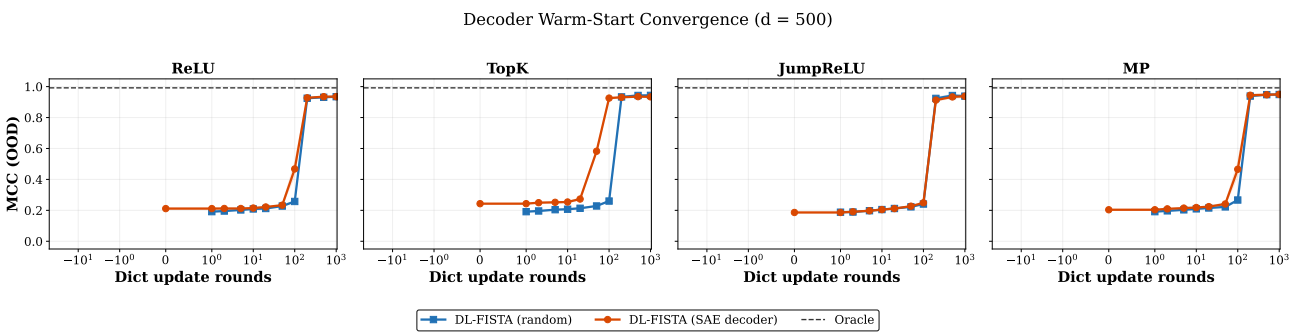

Figure 39: Decoder warm-start convergence, $d_z = 500$.

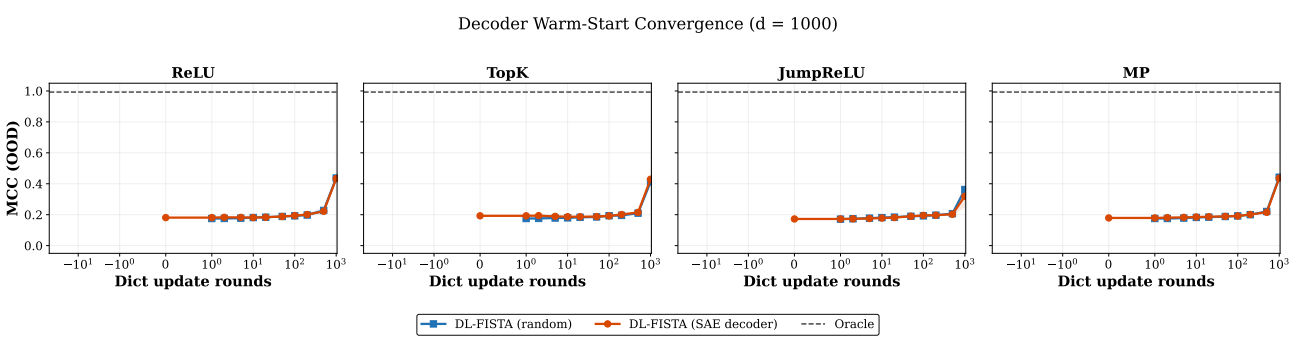

Figure 40: Decoder warm-start convergence, $d_z = 1000$.

Encoder warm-start convergence. The convex-convergence pattern holds across all d_z: cold-start and warm-start reach the same optimum, with warm-starting providing a modest advantage only at very low iteration counts.

Dictionary quality and support recovery. The dictionary quality decomposition and support recovery patterns are consistent across d_z: re-normalising columns never helps, and ReLU/JumpReLU support precision remains catastrophically low.

D.9 PHASE TRANSITION ABLATIONS

Figures 58 to 62 extend the main-text phase transition (Figure 5) with all remaining metrics. The phase transition pattern is consistent: per-sample methods exhibit a sharp transition to near-perfect performance once δ exceeds the compressed-sensing threshold, while SAEs plateau well below across all metrics.